

LABELING & CLASSIFICATION OF BILINGUAL TWEETS

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Abstract—The increasing use of social media platforms has made them a valuable source for gathering data related to public opinion, sentiment analysis and language trends. In the context of bilingual communities, such as those that communicate in both English and Urdu, social media offers a unique opportunity to study the language usage patterns and semantic structures of these communities. The aim of this research is to develop a two phase semi supervised machine learning model to classify bilingual tweets written in English and Roman Urdu based on their content. In the first phase, the model learns on a dataset of human labeled tweets. In the second phase, the model leverages the power of machine learning techniques to automatically categorize tweets into specific categories pertaining to the content in the tweets, such as politics, entertainment, etc. The research will also explore the effectiveness of Multinomial Naive Bayes, Complement Naive Bayes and Linear SVM in achieving accurate and efficient classification. Obtained results showed exceptional results predicting with an accuracy of up to 92%, precision of 89% and recall up to 88%, working with a dataset of approx. 9000 tweets. The results of this research have the potential to contribute to our understanding of the language usage patterns of bilingual communities, and can have multiple different practical applications such as for recognizing trends in marketing or monitoring certain keywords on social media.

Keywords—Machine Learning, Classification Model, Semantics, Sentiment Analysis

I. INTRODUCTION

The research aims to classify bilingual tweets in English and Urdu based on their semantic content using semi-supervised machine learning models. Twitter has become a vital communication platform for people to express their views and opinions in real-time. The study of multilingual content on social media platforms is crucial for various applications such as sentiment analysis, trend analysis, social media monitoring, and opinion mining. The research will focus on identifying and classifying the semantic content of tweets in English and Roman Urdu. The research will use semi-supervised machine learning algorithms to classify bilingual tweets. The dataset will consist of tweets in English and Roman Urdu, which will be pre-processed, tokenized into a dictionary, and labeled. The classification of the tweets will be based on the semantic content, such as keywords, sentiments and topics discussed in the tweet. The study will consider different machine

learning algorithms, such as Support Vector Machines (SVM) [11], Naive Bayes (NB) [11], K-Neighbors Clustering, to classify the tweets.

The research will use a semi-supervised approach to leverage both labeled and unlabeled data. Initially, a small set of tweets will be manually labeled by human annotators to train the machine learning models. The labeled data will then be used to train the machine learning algorithms, which will be used to classify the remaining unlabeled tweets.

The main objective of our research is the rework and improvement of an already existing research [1] centered around the classification of bilingual tweets in English and Roman Urdu into multiclass labels. The main goal was to increase the accuracy of the classification of tweets in the second phase, while also improving the labeling of tweets from the model in the first phase. The main parts that showed major improvement were:

- The dataset, as we expanded the dataset while also reducing the imbalance in the dataset.
- The preprocessing of tweets, as previously tweets were not cleaned and labeled correctly.
- The obtained results, improving the results of tweet labeling and classification.

The tweets are categorized into the following six categories:

- Politics (relating to local and international politics)
- Fun Talks (relating to random conversations as well as any joke or funny posts)
- Technology (relating to any new technological advances or releases)
- Current Affairs (relating to current events, maybe related to different fields such as finance, health, etc.)
- Entertainment (relating to sports, film industry, gaming, etc.)
- Others

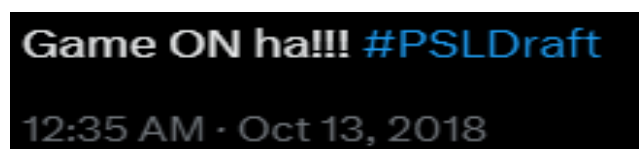


Fig. 1: An example Tweet from the dataset.

The research aims to further understand more about patterns in English and Roman Urdu tweets as previous research on the topic is not very common or very comprehensive. The research aims to contribute to the field of social media analysis by developing a methodology for classifying tweets in English and Urdu, which can be relevant to organizations that rely on social media data for decision-making purposes. The dataset was compiled from tweets that were data mined from Twitter using a script which employs the Twitter API. The script is given hashtags as input [16] that are used by the Twitter API to find any tweets containing or relating to the hashtags specified. This process is then run for hashtags that are popular at different points of time to get variance in the compiled dataset, and to ensure that the tweets are not solely from certain categories or topics.

The original research consisted of a dataset of approximately five thousand tweets. From the five thousand tweets, three thousand tweets were labeled by a human annotator. These were then in the model in the first phase to label the remaining two thousand tweets.

To improve on the results of the research we firstly increased the size of the dataset. From the original five thousand tweets we decided to increase the tweets to approximately nine thousand tweets. From these nine thousand tweets approximately six thousand tweets were human labeled tweets while the remaining three thousand tweets were labeled through the unsupervised model in the first phase of the research. (see more in Section III.B)

Increasing the dataset gave the model more data to learn from in both phases of the research which resulted in improved labeling of the tweets as well as improved results in the classification of the tweets in the latter phase. Increasing the dataset also reduced the risk of the model overfitting to the training dataset and having low testing accuracy which was one of the issues in the previous research.

II. MOTIVATION BEHIND THE RESEARCH

A. Problem Statement

The problem addressed by this research is the lack of an effective methodology for the classification of bilingual tweets in English and Urdu based on their semantic content. Twitter is a widely used social media platform for people to express their views and opinions in real-time, and the analysis of multilingual content on Twitter is crucial for various applications such as sentiment analysis, social media monitoring, and opinion mining. However, the classification of bilingual tweets in English and Urdu poses a challenge due to the lack of labeled data and the complexity of language translation. There being no previously labeled datasets available for Urdu also posed a challenge, meaning that the dataset had to be made from scratch.

Current approaches to classifying bilingual tweets in English and Urdu rely on traditional machine learning algorithms that require a significant amount of labeled data, which is difficult and time-consuming to obtain. Moreover, these approaches do not effectively capture the nuances of the two languages and the interplay between them. Therefore, there is a need for a new methodology that can

effectively classify bilingual tweets in English and Urdu while leveraging unlabeled data.

This research aims to address this problem by developing a semi-supervised machine learning approach to classify bilingual tweets in English and Urdu based on their semantic content. The research will leverage both labeled and unlabeled data and use self-training techniques to improve the accuracy of the classification models. The research will also explore the use of different machine learning algorithms to improve the classification performance. The research's outcome will be a methodology for the semantic classification of bilingual tweets in English and Urdu that can be applied to social media platforms in the future. The research's results will have implications for researchers, marketers, and organizations that rely on social media data for important decision-making purposes or for forecasting purposes.

B. Related Work

Research has been conducted on the bilingual text consisting of English and Roman Urdu but the research is still in its infancy which is proven by there being no definitive or comprehensive reports on the topic.

This research is a continuation of a study conducted previously by Ayaz H. Khan and M. Zubair, [1] which also employs the use of a two-phase approach with the model labeling the tweets in the first phase and classification in the latter phase. The study used a relatively smaller dataset of tweets, with a higher percentage of human-labeled tweets. KNN, Naive Bayes and SVM were the models used for both the labeling as well as the classification. The results obtained showed that Naive Bayes was the most accurate model achieving an accuracy of approximately 87%. However, the labeling in the first phase created an imbalanced dataset with a high number of tweets getting labeled in the "Others" category.

While the study does provide insight into the classification of multilingual tweets using machine learning algorithms, it has also shown that there is still room for further development in this particular subject. Hence, establishing the need for us to further the work done and identify new methodology to improve the semi-supervised machine learning approach to classify bilingual tweets in English and Urdu based on their semantic content.

III. REQUIREMENTS

A. Feature Extraction

To design a model that can effectively and efficiently classify the dataset into appropriate categories some parameters need to be decided for the sake of comparison. For this case involving tweets, the most appropriate parameter would be the keywords [16] used in them. While each tweet may contain an infinite number of word combinations, choosing the appropriate words to constitute a set for comparison remains a difficult task. So, choosing a feature set appropriately is particularly challenging in this regard. So during the labeling process, a set of keywords were compiled specific to each category, for the words appearing most frequently. Some words proved to

be a challenge that could apply to multiple categories and also proved to be difficult to classify in the later stages.

B. Requirement of Data

The machine learning models that are to be used in the research rely on the input of human labeled data that the model can be trained on before making any predictions on future data. As a result, producing a sizable volume of labeled training data may present issues, such as the possibility that each human annotator could assign a different label to a tweet depending on their emotions, culture, country, family background, political or religious opinions, etc. As a result, the representation of labels for training data is diverse and non-concrete, which could negatively impact the classifier's performance.

So for the classification of these a large dataset of appropriate needs to be firstly acquired from and then labeled by human experts before it can be used for training the model. Having a large dataset makes this process have a large overhead in terms of time and manpower, having to go through thousands of tweets to find ones that are appropriate and to find human experts capable of objectively labeling these tweets without bias. For this reason the aspect of unsupervised learning was introduced. The introduction of unsupervised makes it more important that the data is correct as it was the base of learning for any model labeled tweets. To ensure uniformity in the labeling, we ensured that a single human annotator went through the labeled data to make sure that there was no ambiguity in labeling.

IV. LITERATURE REVIEW

This section will give a more detailed explanation of the background of research and the papers used into limitations and advantages.

A. Monolingual

Most of the current research in the field of sentiment analysis is focused on analyzing tweets to determine their sentiment. However, there is a lack of concrete proposed solutions for classifying tweets into multiple classes, which is a major deficiency in the robustness of these models [17, 24]. Additionally, the majority of the research is based on monolingual or structured languages, which makes it difficult to classify tweets that are written in unstructured styles or that use Roman representations. Urdu language, in particular, has received very little attention in this field, comprising less than 1% of the surveyed research articles [23]. The majority of research in this field is focused on Chinese, English, and Arabic, which are the most targeted languages for sentiment analysis. However, some researchers [22] have developed a model that addresses the challenges of classifying Urdu tweets using Roman representation and a multi-lingual approach. This model takes into account the various representations of Urdu, including vowels, and converts

them to Roman Urdu, making it possible to classify them accurately. Overall, there is a need for more research in this field, particularly for non-structured languages like Urdu, to improve the effectiveness and robustness of sentiment analysis models.

B. Pre-Processing

Preprocessing in machine learning refers to the steps taken to prepare data for training a model. Preprocessing is an important step because the quality of the data used to train a machine learning model can have a significant impact on the accuracy and effectiveness of the model. [4, 20, 21, 25] The procedure is further defined in **Section V(A)**.

The first consideration in this research pertains to whether or not to include retweets in the training data. One study [24] suggests that including retweets may add unnecessary weight to the data, while another research [26] suggests that retweets should be included. In our research, we found that including retweets actually improved performance. The second consideration is whether to use hashtags as features for learning [27]. Some researchers have suggested that hashtags can provide a strong indication of a tweet's classification. However, using hashtags as features can lead to biased decisions by the model, such as in the case of a sarcastic text that uses the same hashtag as a positive text. Therefore, including hashtags as features must be approached with caution and consideration of potential biases.

C. Supervised Learning

Supervised learning is a type of machine learning algorithm that involves training a model on a labeled dataset, where each example in the dataset has a known output or target variable. The goal of supervised learning is to learn a mapping between input variables (also known as features or predictors) and output variables. Given the constraint of streaming data, supervised learning models face limitations in handling the continuous and dynamic nature of live data. These models would require constant human intervention to provide labeled data, which can introduce biases. Since the domain of linguistics is involved, biases can arise from influential factors such as political affiliations, emotional influences, and social factors that can impact human decision-making. Consequently, the same text could be assigned different classes by different supervisors, further exacerbating the issue of bias.

D. Semi-supervised Learning

Data augmentation is a method used in semi-supervised learning to generate additional training samples artificially. This technique involves creating synthetic data points to enhance the training process of models. In two research papers, specifically [2] and [28], proposed models have been introduced. These models

employ a filtering mechanism to select elements from unlabelled data that have a high likelihood of being similar to the labeled data. By incorporating these filtered elements, augmented data is generated to augment the training dataset.

E. Unsupervised Learning

Unsupervised learning [18] is a type of machine learning where the model is trained on an unlabeled dataset [26,29,30]. In this domain, several techniques have been employed by researchers [31] to analyze and categorize data. These techniques include term interestingness, topic modeling, and relevance clustering. The primary objective of these approaches is to identify patterns and similarities within the data. Ultimately, it is the algorithm or model's responsibility to differentiate between different data points or cluster similar data together.

F. Algorithms

Among the many machine learning algorithms available, the research employs the use of two variations of Naive Bayes, Multinomial [1,7,11,12,14,15,22,32,35] and Complement [8,13,15], and Linear Support Vector Machine [1,7,9,11,14,15,17,32-35]. These three algorithms proved to be the most effective for the scope of this research. Other algorithms were also tested such as KNN [7,22,33,34], Random Forest [36,37], etc. but provided subpar results.

Algorithm	Multinomial Naive Bayes	Complement Naive Bayes	Linear SVM
Strengths	Fast, efficient for large datasets, good for text classification, handles multi-class problems, robust to noise.	Addresses class imbalance, focuses on misclassified instances, and can improve accuracy.	Effective in high dimensions, models complex boundaries, good generalization, robust to noise.
Weakness	Assumes feature independence, poor with correlated features, limited context capture.	Still assumes feature independence, less effective with correlated features.	Requires careful tuning, slower training, less interpretable.
Suitability	Good for distinct language features, struggles with overlapping vocabularies in bilingual data.	Suitable for imbalanced classes in bilingual contexts, useful when one language dominates.	Strong with mixed language features, captures context well, handles overlapping vocabularies effectively.

Table 1 Comparison of algorithms

V. METHODOLOGY

Our proposed methodology is based on a two phase approach using unsupervised in phase one and supervised learning in phase two.

A. Pre-Processing

Initially, the data is converted to plain text from JSON, then the tweet tag of the JSON is extracted removing all other JSON tags, links & emoticons. To eliminate any spelling ambiguity and duplicate feature extraction, the text is converted to lowercase. After that all stop words such as 'is', 'am', 'are', 'they', etc. are removed as they hold no semantic value and are only used for the structuring of sentences. Following that, all user references such as @TwitterUser and #Hashtags are removed. Keeping hashtags could act as a biased feature that can cause misleading classifications rather than the actual semantic classification of the text itself. The entire text is then run through an encoding checker to ensure that every character has been accurately transformed before all non-alphabetic characters are eventually eliminated.

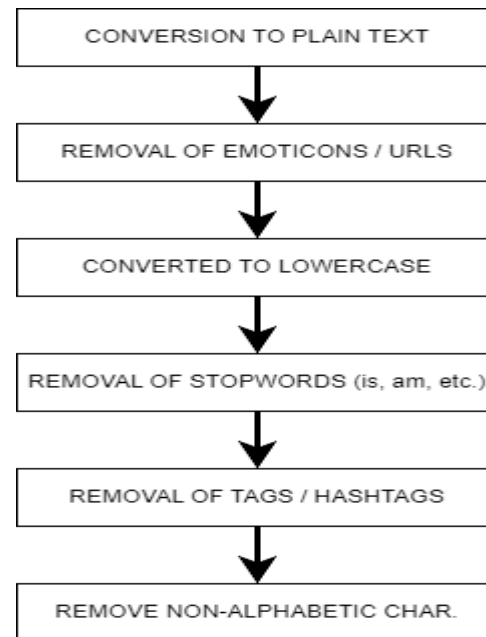


Fig. 2.1 The flow of preprocessing

B. Feature Selection

Feature selection is the process of selecting a subset of relevant features from the original dataset to be used in building a machine learning model. The primary concept behind feature selection is that data may contain some redundant or unnecessary features that may be removed without any loss of important data. By selecting only the most relevant features, we can improve the accuracy and efficiency of the model, reduce the risk of overfitting, and make the model easier to interpret.

For dictionary generation [19] we used most frequently appearing words in tweets and applied a Bag of Words approach.[5] Using one hot encoding, words in tweets are converted into binary representation such that each word occurring in the feature set is represented as 1 and the ones that do not as 0.

A similarity index was also implemented during the dictionary generation to make sure words with multiple variations (such as hello, helo) would be interpreted as a single entry in the dictionary. However, the implementation of a similarity index had problems with the time complexity being very high. The problem would only increase as the size of the dictionary would increase as each word would be compared to every word in the dictionary to find similarities. The process of comparing similarities could take up to 30 minutes even for the smaller dictionary sizes. The results of implementing the similarity index did not warrant the trade-off in performance of the model, as the improvements were negligible to non-existent in most cases.

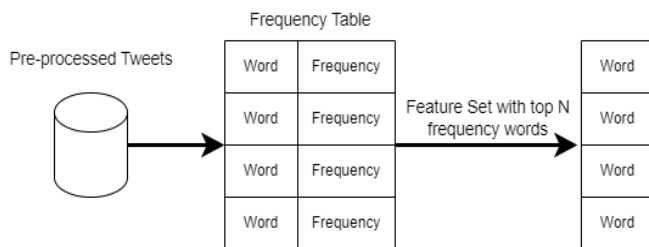


Fig. 2.2 Process for Dictionary generation

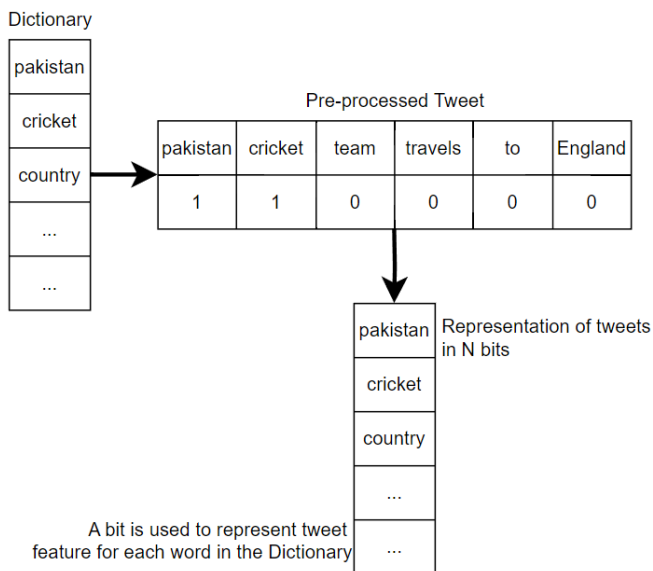


Fig. 2.3 Tweet Vectorization using One Hot Encoding

C. Models

Research based on the classification of bilingual tweets using unsupervised and supervised machine learning models can leverage a variety of algorithms for classification. So to find an algorithm that could provide the most accurate and efficient was of utmost importance. So

the following algorithms were chosen as they provided the best results from the many models used in testing.

1) Multinomial Naive Bayes:

Multinomial Naive Bayes [12,14,15] is a probabilistic algorithm used for classification tasks. It is a variant of the Naive Bayes algorithm, which assumes that the features are independent of each other given the class.[7] The input data is represented as a matrix of features, where each row represents a document and each column represents a feature, such as the frequency of a word. The target variable is represented as a vector of labels. The algorithm estimates the prior probabilities of each class based on the frequency of the labels in the training data. It also estimates the conditional probabilities of each feature given the class based on the frequency of the feature in the documents of each class. Model prediction: Given a new document, the algorithm calculates the posterior probabilities of each class given the features of the document using Bayes' theorem (refer to equation 4.1). It then selects the class with the highest posterior probability as the predicted class.

The key assumption of the Multinomial Naive Bayes algorithm is that the features are independent given the class. This assumption allows the algorithm to estimate the conditional probabilities of each feature independently, which simplifies the calculation of the posterior probabilities.

Despite its simplicity and assumptions, Multinomial Naive Bayes has been shown to be effective in many text classification tasks, especially when the number of features is large and the amount of training data is limited.

$$\text{Bayes Theorem: } P(A|B) = \frac{P(A) P(B|A)}{P(B)} \quad (1)$$

2) Complement Naive Bayes:

Complement Naive Bayes [13,15] is a variant of the Naive Bayes algorithm used for classification tasks. Unlike the standard Naive Bayes algorithm, which assumes that the features are independent of each other given the class, the Complement Naive Bayes algorithm takes into account the complement of the class. The Complement Naive Bayes algorithm is particularly useful for imbalanced datasets where one class has a much larger number of instances than other classes. In such cases, the standard Naive Bayes algorithm tends to favor the majority class and perform poorly on the minority class. The Complement Naive Bayes algorithm addresses this problem by modeling the complement of the class instead of the class itself. [8]

The data given as input is a matrix of features same as for the Multinomial variation of the Naive Bayes algorithm. The algorithm calculates the posterior probability based on the frequency of labels and conditional probabilities of each

feature given the complement of the class based on the frequency of the feature in the documents that do not belong to the class. Finally, for the prediction, it calculates probability using the Bayes theorem (refer to equation 4.1) for all the classes. The class with the lowest probability is selected as it represents the highest chance of being in that class.

The Complement Naive Bayes algorithm works by balancing the contribution of each feature to the classification of the classes. Features that are common in the complement of the class are weighted more heavily than features that are common in the class itself. This helps the algorithm to better capture the discriminative features of the minority class.

Despite its effectiveness in handling imbalanced datasets, the Complement Naive Bayes algorithm has higher computational complexity than the standard Naive Bayes algorithm and may require more training data to achieve good performance.

3) *Linear Vector Support Machine (Linear SVM):*

Linear Support Vector Machine (SVM) [11,14,15,17] is an algorithm that is widely used for classification tasks. It is a linear model that works by finding the hyperplane that separates the data points into different classes. [9]

A matrix of features is passed as input into the model similar to the previous models and a vector of labels is given as the result. The algorithm finds the hyperplane that maximizes the margin between the closest data points of different classes. The margin is the distance between the hyperplane and the closest data points, and maximizing it helps to reduce the risk of overfitting. The hyperplane is represented by a weight vector w and a bias term b , and is learned by minimizing a loss function that penalizes misclassifications. Given a new data point, the algorithm calculates the distance between the data point and the hyperplane. If the distance is positive, the data point is classified as belonging to one class, and if the distance is negative, it is classified as belonging to the other class.

Linear SVM works well when the data points are linearly separable, that is, when it is possible to draw a straight line (or hyperplane in higher dimensions) that separates the data points into different classes. In cases where the data points are not linearly separable, a variant of SVM called the kernel SVM can be used. The kernel SVM maps the data points into a higher-dimensional space where they can be linearly separated.

Linear SVM proved to be the most computationally demanding model of the three used in the project. The processing time increased considerably using Linear SVM, especially if the model was used in both phases and with high dictionary size, as is mentioned later.

D. *Unsupervised Learning (PHASE 1)*

The models previously used for the first phase were the following:

- K-Neighbors Classifier
- Solid Vector Machine
- Multinomial Naïve Bayes

The mentioned provided varying results with the Multinomial Naïve Bayes proving to be the best out of the 3. However, all 3 models exhibited signs of bad labeling, meaning that the majority of the tweets were labeled into the category of “others”, which after human inspection proved to be the incorrect label more often than not. Upon further inspection of the dataset, it was found that the dataset was imbalanced meaning that the tweets in certain categories heavily outnumbered the tweets in the remaining categories. Almost half of the tweets in the original human labeled dataset belonged to either the “current affairs” category or “others” category. Different models were used to better tackle the mentioned issues and have better unsupervised labeling for the dataset.

E. *Supervised Learning (PHASE 2):*

The second phase of the research consists of predicting correct labels for the tweets. The dataset now consists of both the approx. six thousand human labeled tweets as well as the three thousand tweets labeled by the model used in the first phase. [10]

The results obtained from the previous research went as high as 87.5% accuracy when using Multinomial Naïve Bayes for both the phases, while using SVM also provided an accuracy crossing 80%. While a high accuracy was obtained there was the downside of having an imbalanced dataset, that had been further imbalanced by the addition of the tweets labeled in the first phase, meaning that the accuracy did not provide the best picture of the outcome.

In this version of the research, measures were already taken to fix the problem of the imbalanced dataset through increasing the size of the dataset and using specific models that worked better against imbalanced datasets. For this phase, we also tuned the hyperparameters until we were able to find the best results. For hyperparameter tuning, we first identified the necessary parameters and made a HP space. From the space, we conducted a Manual Search for the parameters that provided the best results. The following models and parameters were used:

- ComplementNB(norm=True)
- MultinomialNB(alpha=0.001)
- svm.LinearSVC(C=12.0, max_iter = 7000, random_state=42)

1. *VI Performance Evaluation Metrics:*

To assess the efficacy of the suggested model, we've utilized the subsequent performance measures. The metrics are shown in terms of:

-**True Positive (TP)**, these are the classifications that are positive and were predicted as positive.
-**True Negative (TN)**, these are the classifications that are negative and were predicted as negative.
-**False Positive (FP)**, these are the classifications that are negative but were predicted as positive.
-**False Negative (FN)**, these are the classifications that are positive but were predicted as negative.

1. Accuracy shows the total number of tweets that were correctly classified compared to the total number of classifications. It is calculated by the following equation:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \quad (2)$$
2. Recall is the percentage of classification that were correctly classified as positive compared to the total number of positive tweets. Recall is determined as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100 \quad (3)$$

3. Precision is the percentage of classifications correctly classified as positive with respect to the total number of positive classifications. It can determine by:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100 \quad (4)$$
4. VVVF-Measure is represented as the balance between the Precision and Recall of the classification experiment and is used in determining the efficiency of the test's accuracy. It can be calculated using:

$$\text{F-Measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100 \quad (5)$$

VI. RESULTS

1. Results of Phase 1:

A. Using Multinomial Naive Bayes (Default Parameters):

a) Dictionary Size: 3000 Words:

Multinomial Naive Bayes		Percentage
Politics	218	7.14
Fun Talk	131	4.29
Technology	8	0.26
Current Affairs	1414	46.33
Entertainment	69	2.26
Others	1212	39.71

Table 2.1: Results of Multinomial NB with 3000-word dictionary

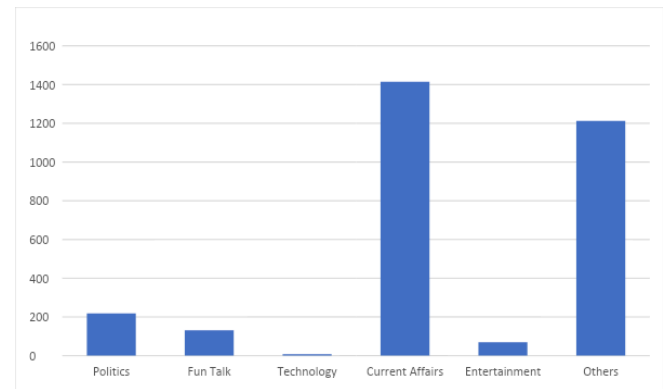


Fig. 3.4: Distribution of labels from Multinomial NB (3000 words)

Performing the first phase using Multinomial Naive Bayes with a dictionary of 3000, with 86% of the tweets being labeled as Current Affairs and Others.

b) Dictionary Size: 3500 Words:

Multinomial Naive Bayes		Percentage
Politics	226	7.40
Fun Talk	120	3.93
Technology	11	0.36
Current Affairs	1415	46.36
Entertainment	61	1.99
Others	1219	39.94

Table 2.2: Results of Multinomial NB with 3500-word dictionary

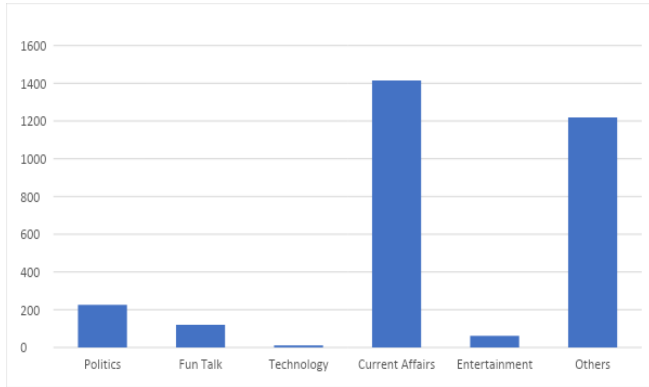


Fig. 3.5: Distribution of Labels from Multinomial NB (3500 words)

Using a dictionary size of 3500 shows similar results with approx. 86% of tweets being labeled as either “Current Affairs” or “Others”. The other categories saw little to no variance.

c) Dictionary Size: 4000 Words:

Multinomial Naive Bayes		Percentage
Politics	217	7.11
Fun Talk	113	3.70
Technology	8	0.26
Current Affairs	1436	47.05
Entertainment	56	1.83
Others	1222	40.03

Table 2.3: Results of Multinomial NB with 4000-word dictionary

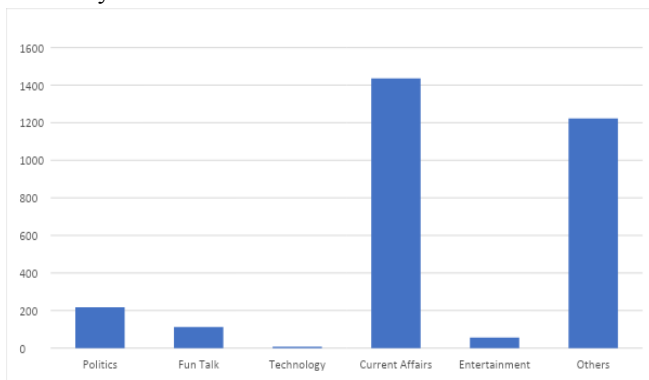


Fig. 3.6: Distribution of labels from Multinomial NB (4000 words)

With a dictionary size of 4000, 87% of tweets were labeled as “Current Affairs” or “Others”. In conclusion for Multinomial Naive Bayes, the dictionary size didn’t affect the label distribution.

B. Using Complement Naive Bayes (Default Parameters):

a) Dictionary Size: 3000 Words:

Complement Naive Bayes		Percentage
Politics	521	17.07
Fun Talk	231	7.57
Technology	77	2.52
Current Affairs	1286	42.13
Entertainment	190	6.23
Others	747	24.48

Table 2.4: Results of Complement NB with 3000-word dictionary

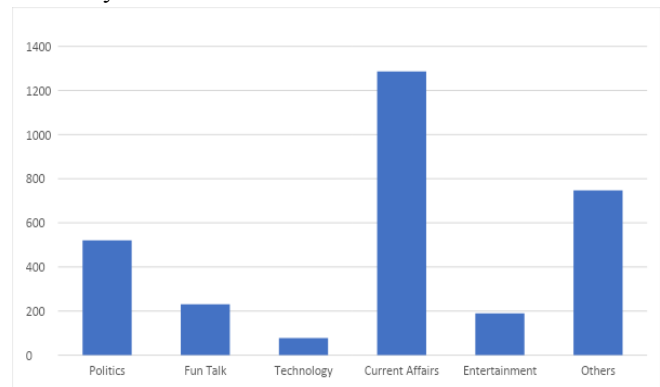


Fig. 3.7: Distribution of labels from Complement NB (3000 words)

Using Complement Naive Bayes for the first phase with a 3000 word dictionary, approx. 84% of the tweets were labeled into “Current Affairs”, “Others” or “Politics”.

b) Dictionary Size: 3500 Words:

Complement Naive Bayes		Percentage
Politics	472	15.46
Fun Talk	212	6.95
Technology	65	2.12
Current Affairs	1322	43.32
Entertainment	187	6.13
Others	794	26.02

Table 2.5: Results of Complement NB with 3500-word dictionary

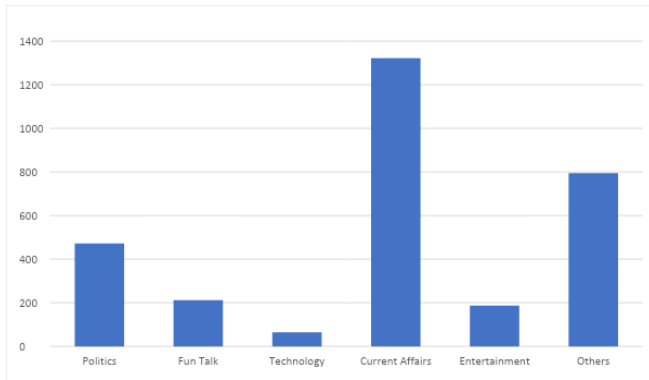


Fig. 3.8: Distribution of labels from Complement NB (3500 words)

With a 3500 word dictionary, the results are similar with 84% tweets being labeled into the three categories. But “Politics” saw a slight decrease while the opposite was the case for “Others”.

c) Dictionary Size: 4000 Words:

Complement Naive Bayes		Percentage
Politics	457	14.97
Fun Talk	198	6.48
Technology	64	2.10
Current Affairs	1343	44
Entertainment	172	5.64
Others	818	26.80

Table 2.6: Results of Complement NB with 4000-word dictionary

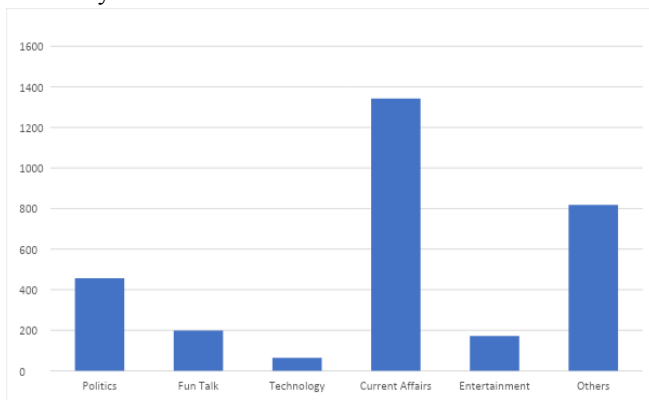


Fig. 3.9: Distribution of labels from Complement NB (4000 words)

For a dictionary with 4000 words, 86% of the tweets were labeled into the three previously mentioned categories. In conclusion, the tweets were mostly distributed in three categories, with the number of tweets increasing in “Others” when the dictionary size was increased.

C. Using Linear SVM (Default Parameters):

a) Dictionary Size: 3000 Words:

Linear SVM		Percentage
Politics	210	6.88
Fun Talk	213	6.98
Technology	22	0.72
Current Affairs	1198	39.2
Entertainment	106	3.47
Others	1303	42.69

Table 2.7: Results of Linear SVM with 3000-word dictionary

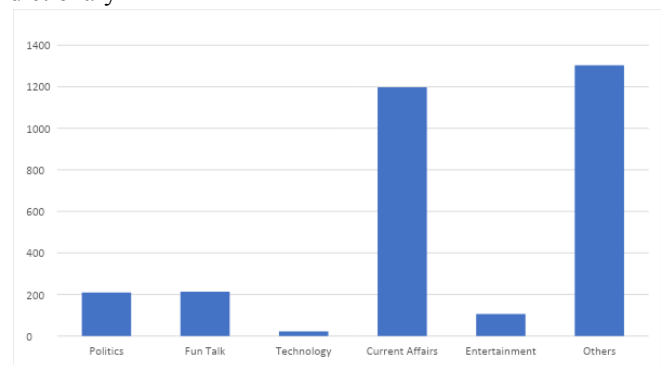


Fig. 3.10: Distribution of labels from Linear SVM (3000 words)

Using Linear SVM with a 3000 word dictionary, approx. 82% were classified into “Current Affairs” or “Others”. “Politics” and “Fun Talk” also had approx. 7% of labeled tweets each.

b) Dictionary Size: 3500 Words:

Linear SVM		Percentage
Politics	214	7.01
Fun Talk	207	6.78
Technology	21	0.69
Current Affairs	1175	38.50
Entertainment	102	3.34
Others	1333	43.68

Table 2.8: Results of Linear SVM with 3500-word dictionary

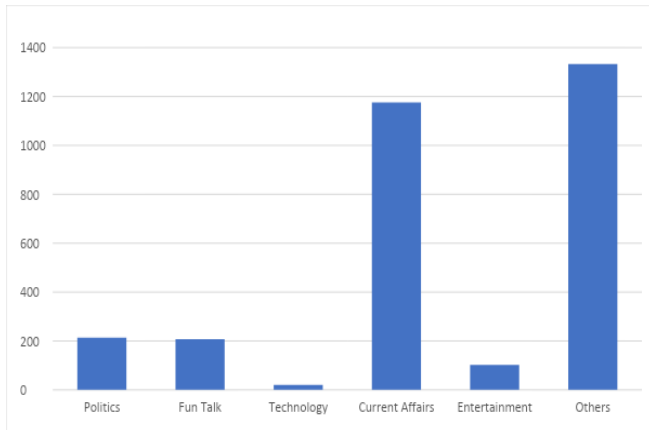


Fig. 3.11: Distribution of labels from Linear SVM (3500 words)

With a dictionary size of 3500 words, the results were similar with 82% of the tweets being labeled as “Current Affairs” or “Others”, and 7% each for “Politics” and “Fun Talk”.

c) Dictionary Size: 4000 Words:

Linear SVM		Percentage
Politics	209	6.85
Fun Talk	198	6.49
Technology	22	0.72
Current Affairs	1179	38.63
Entertainment	106	3.47
Others	1338	43.84

Table 2.9: Results of Linear SVM with 4000-word dictionary

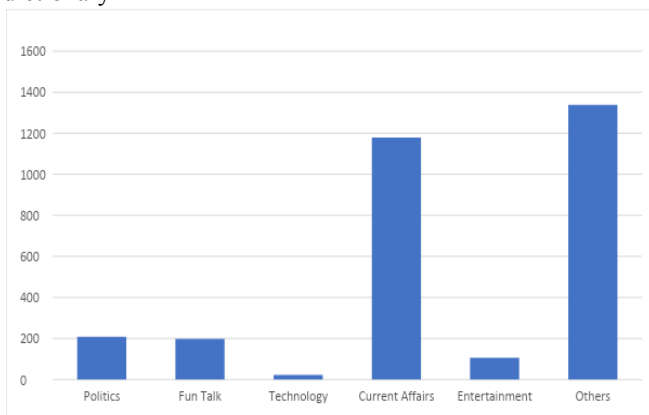


Fig. 3.12: Distribution of labels from Linear SVM (4000 words)

Using a 4000 word dictionary yielded the same results as the two previous dictionary sizes. In conclusion, the dictionary size did not have much of an effect on the labeling of the tweets.

D. Conclusion of Phase 1:

After using three different models along with varying the dictionary, we obtained the data shown above. It shows the imbalance in effect with the Multinomial Naïve Bayes and Linear SVM models categorizing around 80% to 85% of the 3052 unlabeled tweets into either current affairs category or the others category. Whereas, the Complement Naïve Bayes had a more balanced distribution of labels showcasing that this model is indeed more effective in handling imbalanced datasets, making it the best fit for the unsupervised phase of the research.

2. Results of Phase 2:

The results of the second phase while varying the dictionary size are as follows:

Dictionary Size: 3000 words

		1st Phase			
		Measures	Linear SVM	Complement NB	Multinomial NB
2nd Phase	Linear SVM	Accuracy	0.9	0.84696	0.86022
		Recall	0.82416	0.77351	0.81909
		Precision	0.85108	0.80175	0.80710
		F-Measure	0.83641	0.78588	0.8122
	Complement NB	Accuracy	0.67624	0.73480	0.72872
		Recall	0.62199	0.68439	0.67522
		Precision	0.53512	0.60089	0.56781
		F-Measure	0.56005	0.62921	0.60067
	Multinomial NB	Accuracy	0.79171	0.79116	0.83535
		Recall	0.71930	0.71899	0.78564
		Precision	0.72558	0.74950	0.78046
		F-Measure	0.72151	0.73263	0.78233

Table 2.10: Results of Supervised Learning with 3000 word dictionary

Dictionary Size: 3500 words

		1st Phase			
		Measures	Linear SVM	Complement NB	Multinomial NB
2nd Phase	Linear SVM	Accuracy	0.90607	0.85082	0.86961
		Recall	0.86675	0.82002	0.80821
		Precision	0.87632	0.79188	0.79613
		F-Measure	0.87127	0.80244	0.80101
	Complement NB	Accuracy	0.69281	0.76243	0.71602
		Recall	0.66765	0.71640	0.66156
		Precision	0.55309	0.63788	0.56616
		F-Measure	0.58277	0.66516	0.59698
	Multinomial NB	Accuracy	0.78895	0.80828	0.80552
		Recall	0.74638	0.74980	0.73984
		Precision	0.74358	0.77153	0.77218
		F-Measure	0.74175	0.75834	0.75465

Table 2.11: Results of Supervised Learning with 3500-word dictionary

Dictionary Size: 4000 words

		1st Phase			
		Measures	Linear SVM	Complement NB	Multinomial NB
2nd Phase	Linear SVM	Accuracy	0.92430	0.85469	0.86464
		Recall	0.88740	0.80109	0.78752
		Precision	0.89427	0.81790	0.81980
		F-Measure	0.88876	0.80315	0.80230
	Complement NB	Accuracy	0.71602	0.76519	0.72707
		Recall	0.68578	0.69510	0.66267
		Precision	0.59245	0.63485	0.57807
		F-Measure	0.61946	0.65881	0.60728
	Multinomial NB	Accuracy	0.81160	0.79502	0.83204
		Recall	0.77578	0.73724	0.75933
		Precision	0.77023	0.75034	0.79786
		F-Measure	0.77062	0.74243	0.77621

Table 2.12: Results of Supervised Learning with 4000-word dictionary

A. Outcome of Phase 2:

The above data presents the results of the second phase of the research, which utilized three different models and three variations in the dictionary size. The data analysis provides several noteworthy findings. The foremost observation is that Linear SVM is the optimal model to use in both phases of the research as it achieves a consistent 90% accuracy, with a maximum of 92%. The success of Linear SVM can be attributed to the model being able to generalize well in high dimension spaces such as text and the data having clearly defined classes. The drawback being that Linear SVM in both phases can have high computational time especially with the increased dataset. Additionally, employing Linear SVM in the second phase alongside the two Naïve Bayes models yields decent results, with accuracy levels averaging around 85% across all cases

and other performance metrics demonstrating good results. On the other hand, utilizing Complement Naïve Bayes in the second phase does not generate comparable outcomes as the previous model, with accuracy rates hovering around 70%. The Multinomial Naïve Bayes model shows mixed results, ranging from 78% to 83%.

The results above also indicate that increasing the size of the dictionary has either a positive effect or no effect on the accuracy of the outcome in nearly all cases. Since there were no negative impacts, it is suggested to use a dictionary size of 4000 words as it yielded the best results. While expanding the dictionary size beyond 4000 words had a negligible effect on the outcome, it also heightened the computational time for the models especially in the case of Linear SVM, a model that is slower when given larger sets of data.

VII. CONCLUSION

After analyzing the results of both phases of the research, we have determined that the best combination of models is Complement Naïve Bayes in the first phase and Linear SVM in the second phase. This combination provides an optimal trade-off between correct labeling and high classification accuracy. However, there are still opportunities to improve the model, with a larger dataset being a possibility as well as diversifying the categories for classification. Another possible future addition can be assigning multiple labels to a single tweet. The addition of more categories and multi labeled tweets will reduce any ambiguity there may be in the classified tweets, as some tweets may fall into different categories depending on the annotator or could be relevant to multiple categories. Increasing the size of the dataset will improve learning for the model. However, the additions will increase the execution time as all three cases will require more computational power.

Overall, the research has yielded promising results, with the chosen combination of Complement Naïve Bayes and Linear SVM achieving an accuracy of 85%. The use of Linear SVM in both phases was particularly effective, with classification accuracy reaching as high as 92% with a slight trade-off in the accuracy of the first phase. These results improve on the results of the previous study, using it as a base to improve the overall classification of tweets in the first phase which in turn led to more accurate results in the latter phase of the project.

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